Package 'levyBridge'

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Title Stochastic Bridges for Levy Degradation Processes

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Description Exact simulation of the bridges of Levy degradation processes (gamma, Wiener, inverse Gaussian, variance-gamma and Meixner processes) and resampling of trajectories observed at irregular inspection times onto a regular time grid, following the interpolation algorithm of Grall-Maes (2012) <doi:10.1177/1748006X12462522>. The reconstructed increments are the basis of goodness-of-fit tests.

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`levyBridge-package` *levyBridge: stochastic bridges for Lévy degradation processes*

Description

`levyBridge` resamples degradation trajectories observed at irregular inspection times onto a regular time grid, by exact simulation of the bridge of the process between the known values. Under the model, the reconstructed increments are independent and identically distributed: they are the input of the goodness-of-fit tests for Lévy degradation processes.

Bridge of a Lévy process

Let X be a Lévy process such that X_s has the density f_s, and $t_A < t < t_B$. Given $X_{t_A} = x_A$ and $X_{t_B} = x_B$, X_t has the density

$$f_{t|t_A,t_B}(x) = \frac{f_{t-t_A}(x - x_A)\, f_{t_B-t}(x_B - x)}{f_{t_B-t_A}(x_B - x_A)}$$

(Hoyle, Hughston and Macrina, 2011). Each `_bridge` function draws exactly from this law, for one process:

Function	Process	Parameters	Simulation of the bridge
`gamma_bridge()`	gamma	shape `a`	beta variable
`wiener_bridge()`	Wiener	`sigma`	Gaussian variable
`ig_bridge()`	inverse Gaussian	`eta`	IG or scaled inverse IG
`vg_bridge()`	variance-gamma	`theta`, `sigma`, `nu`	GIG, beta, Gaussian
`meixner_bridge()`	Meixner	`a`, `d`, `m`	rejection, Cauchy envelope

The law of a bridge never depends on all the parameters of the process: the gamma bridge does not depend on the rate `b`, the Brownian bridge does not depend on the drift `mu`, the inverse Gaussian bridge does not depend on `mu`, and the Meixner bridge does not depend on `b`. Click on a function name for the law of its bridge and the details of the simulation.

Resampling methods

Every `_bridge` function has an argument `method`:

- `"Bridge"` (default): interpolation algorithm of Grall-Maës (2012); see also the help page of `gamma_bridge()`. A trajectory observed at $0 = t_0 < t_1 < \cdots < t_N$ is resampled on the regular grid $t'_k = k\, t_N/N$, $k = 0, \ldots, N$: as many points as observations, with the time step $\Delta t_{eq} = t_N/N$. The grid is covered from left to right; each grid point is drawn from the bridge between the last known value before it (an observation or the previous grid point, already drawn) and the next observation. A grid point equal to an inspection time keeps the observed value;
- `"ExtBridge"`: extension of this method whose grid contains all the inspection times. Not available yet.

Data

All the `_bridge` functions accept the same formats:

- **one trajectory**: a vector of levels `x` and a vector of inspection times `t`, starting at 0, or a data frame whose first column contains the times and second column the levels;
- **several trajectories with the same inspection times**: the matrix returned by the `levysim` simulation functions, with columns `t`, `X1`, `X2`, ...;
- **several trajectories, each with its own inspection times**: a list of data frames, one per trajectory (first column the times, second column the levels), for instance the list returned by `levysim` with one plan per trajectory.

Each trajectory gets its own regular grid and its own time step.

Results

A `_bridge` function returns an object of class `"levybridge"`:

- `res` prints the model, the parameters, the time steps and the first resampled values;
- `res[[k]]` is the k-th trajectory on its regular grid, a data frame with columns `t` and `X`;
- `attr(res, "teq")` gives the time step of the grid of each trajectory;
- `increments()` gives the reconstructed increments, one row per increment with the columns `id` (trajectory), `teq` and `dx`: they are the values to be tested;
- `plot_bridge()` compares the observed and resampled values (`ggplot2`).

The bridge values are random: use `set.seed()` before a resampling to make it reproducible.

Goodness-of-fit test, step by step

The package covers steps 2 to 4 of the methodology of goodness-of-fit tests for a parametric family of Lévy processes:

1. **Estimate** the parameters from the observed trajectories, for instance with the package `levyfit` (`gamma_mle()`, `meixner_mle()`, ...);
2. **Law of the bridge**: it is given in the help page of each `_bridge` function;
3. **Simulation of the bridge**: done by the `_bridge` function, with the estimated parameters;
4. **Pseudo-sample on a regular grid**: the result of the `_bridge` function, with the method "Bridge";
5. **Choose a test** for an i.i.d. sample of the law of $X_{\Delta t_{eq}}$, for instance Kolmogorov-Smirnov, Cramér-von Mises or Anderson-Darling;
6. **Apply the test** to the reconstructed increments `increments(res)$dx`.

Author(s)

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References

Grall-Maës, E. (2012). Use of the Kolmogorov-Smirnov test for gamma process. *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, 226(6), 624-634. doi:10.1177/1748006X12462522

Hoyle, E., Hughston, L. P. and Macrina, A. (2011). Lévy random bridges and the modelling of financial information. *Stochastic Processes and their Applications*, 121(4), 856-884.

Tito, A. R. and Paroissin, C. (in preparation).

Examples

```
# 1. One gamma trajectory observed at irregular times
t <- c(0, 1.2, 2, 3.8, 5.5, 7.1, 10.4, 12.9, 15.3, 18.7, 22.5)
set.seed(1)
x <- c(0, cumsum(rgamma(10, shape = 2 * diff(t), rate = 3)))

# Resampling on the regular grid 0, 2.25, 4.5, ..., 22.5
r <- gamma_bridge(x, t, a = 2)
r
r[[1]]
attr(r, "teq")
increments(r)
plot_bridge(r)

# 2. The same trajectory with the other processes (parameters of the bridge only)
wiener_bridge(x, t, sigma = 1.5)[[1]]
ig_bridge(x, t, eta = 3)[[1]]
vg_bridge(x, t, theta = 0.5, sigma = 1.5, nu = 2)[[1]]
meixner_bridge(x, t, a = 1, d = 1.5, m = 0.2)[[1]]

# 3. Several trajectories, each with its own inspection times (a list)
liste <- list(data.frame(t = c(0, 1, 3, 6), X = c(0, 0.4, 2.3, 2.9)),
              data.frame(t = c(0, 2, 4, 5, 9), X = c(0, 1.6, 2.8, 3.1, 4.2)))
r2 <- gamma_bridge(liste, a = 2)
attr(r2, "teq")
increments(r2)

# 4. Complete use: simulation (levysim), estimation (levyfit), bridge
plan <- levysim::inspection_times("increasing", n = 20, n_traj = 10, tmax = 40)
sims <- levysim::sim_gamma(plan, a = 2, b = 3)

# Step 1: estimation of the parameters
est <- levyfit::gamma_mle(sims)
est

# Steps 3 and 4: resampling with the estimated shape
r3 <- gamma_bridge(sims, a = est["a"])
plot_bridge(r3, trajectory = 1:4)

# Step 6: the reconstructed increments, to be tested against the law
# Gamma(a teq, b) with the estimated parameters
d <- increments(r3)
head(d)
```

`gamma_bridge` *Gamma bridge: resampling on a regular grid*

Description

Resamples one or several trajectories of a homogeneous gamma process, observed at irregular inspection times, onto a regular time grid, by exact simulation of the gamma bridge between the known values.

Usage

```
gamma_bridge(x, t = NULL, a, method = "Bridge")
```

Arguments

`x`	Degradation levels of one trajectory; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels (for instance the result of a `levysim` simulation function).
`t`	Inspection times of the trajectory, starting at 0 and strictly increasing. Not used when `x` is a list.
`a`	Shape parameter of the gamma process, $a > 0$, for instance estimated with the function `gamma_mle()` of the package `levyfit`.
`method`	`"Bridge"` (default) for the algorithm of Grall-Maës (2012), or `"ExtBridge"` for its extension (not available yet).

Value

An object of class `"levybridge"`: a list with one data frame per trajectory, with columns `t` (regular grid) and `X` (resampled levels). Its attributes are `model`, `parameters`, `method`, `teq` (time step of the grid of each trajectory) and `observed` (the observed data, one data frame per trajectory). Use `increments()` for the reconstructed increments and `plot_bridge()` to compare observed and resampled values.

Model

The homogeneous gamma process with shape $a > 0$ and rate $b > 0$ has $X_0 = 0$, independent increments and $X_t - X_s \sim \Gamma(a(t-s), b)$ for $s < t$ (shape $a(t-s)$, rate b). Its trajectories are increasing.

Law of the bridge

Let $t_A < t < t_B$, $\tau_1 = t - t_A$, $\tau_2 = t_B - t$ and $\Delta = x_B - x_A$. Given $X_{t_A} = x_A$ and $X_{t_B} = x_B$,

$$X_t = x_A + \Delta\, Y, \qquad Y \sim \mathrm{Beta}(a\tau_1, a\tau_2).$$

The law of the bridge does not depend on the rate b: only the shape a is needed. When $\Delta = 0$ (two equal observed values, which can happen after rounding), the bridge is constant.

Resampling algorithm (method "Bridge")

This is the interpolation algorithm of Grall-Maës (2012), used for all the processes of the package. Let $0 = t_0 < t_1 < \cdots < t_N$ be the inspection times of a trajectory and $x_0, \ldots, x_N$ the observed levels. The regular grid is $t'_i = i\,\Delta t_{eq}$, $i = 0, \ldots, N$, with $\Delta t_{eq} = t_N/N$: it has as many points as observations and the same end points.

1. $x'_0 = x_0$ and $x'_N = x_N$.
2. For $i = 1, \ldots, N-1$, in increasing order:
   - if t'_i is an inspection time, x'_i is the observed value;
   - otherwise, x'_i is drawn from the bridge between the nearest known point on the left (an inspection or the grid point t'_{i-1} already drawn, whichever is closer) and the nearest inspection on the right.

By the Markov property of the bridge, the vector $(x'_1, \ldots, x'_{N-1})$ follows exactly the conditional law of $(X_{t'_1}, \ldots, X_{t'_{N-1}})$ given the observations. Hence, if the data come from the model and the parameters are the true ones, the reconstructed increments $x'_i - x'_{i-1}$ (see `increments()`) are independent with the law of $X_{\Delta t_{eq}}$: they can be tested with an i.i.d. goodness-of-fit test. A grid point that equals an inspection time up to rounding errors (relative difference below 10^{-9}) is set to that inspection time.

Method "ExtBridge"

Extension of the method of Grall-Maës that integrates the observations. It is not available yet: `method = "ExtBridge"` stops with a message.

References

Grall-Maës, E. (2012). Use of the Kolmogorov-Smirnov test for gamma process. *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, 226(6), 624-634. doi:10.1177/1748006X12462522

Avramidis, A. N., L'Ecuyer, P. and Tremblay, P.-A. (2003). Efficient simulation of gamma and variance-gamma processes. *Proceedings of the 2003 Winter Simulation Conference*, 319-326.

See Also

- `increments()`: the reconstructed increments;
- `plot_bridge()`: observed and resampled values;
- `wiener_bridge()`: bridge of the Wiener process;
- `ig_bridge()`: bridge of the inverse Gaussian process;
- `vg_bridge()`: bridge of the variance-gamma process;
- `meixner_bridge()`: bridge of the Meixner process.

Examples

```
# 1. One trajectory observed at 11 irregular times (a = 2, b = 3)
t <- c(0, 1.2, 2, 3.8, 5.5, 7.1, 10.4, 12.9, 15.3, 18.7, 22.5)
set.seed(1)
x <- c(0, cumsum(rgamma(10, shape = 2 * diff(t), rate = 3)))
set.seed(1)
r <- gamma_bridge(x, t, a = 2)
r

# The resampled trajectory, on the regular grid 0, 2.25, 4.5, ..., 22.5
r[[1]]

# Time step of the grid: t_N / N = 22.5 / 10
attr(r, "teq")

# 2. Reconstructed increments, the input of the goodness-of-fit tests
increments(r)

# 3. Observed and resampled values
plot_bridge(r)

# 4. Grid points equal to inspection times keep the observed values:
#    here the grid is 0, 2, 4, 6, 8, 10 and 2 and 6 are inspection times
```

```
t2 <- c(0, 2, 3, 6, 7, 10)
x2 <- c(0, 1.4, 2.0, 4.1, 4.6, 6.8)
gamma_bridge(x2, t2, a = 2)[[1]]

# 5. One trajectory given as a data frame (times, then levels)
gamma_bridge(data.frame(t = t2, X = x2), a = 2)

# 6. Several trajectories, each with its own inspection times:
#    one regular grid and one time step per trajectory
liste <- list(data.frame(t = c(0, 1, 3, 6), X = c(0, 0.8, 2.1, 4.3)),
              data.frame(t = c(0, 2, 4, 5, 9), X = c(0, 1.2, 2.9, 3.1, 6.2)))
r2 <- gamma_bridge(liste, a = 2)
r2
attr(r2, "teq")
increments(r2)
plot_bridge(r2, trajectory = 1:2)

# 7. The bridge values are random: set.seed() makes them reproducible
set.seed(2); y1 <- gamma_bridge(x, t, a = 2)[[1]]$X
set.seed(2); y2 <- gamma_bridge(x, t, a = 2)[[1]]$X
identical(y1, y2)

# 8. Equal observed values (after rounding) give a constant bridge,
#    a decrease is impossible for this increasing process
gamma_bridge(c(0, 1, 1, 2), c(0, 1, 2.5, 3), a = 2)[[1]]
try(gamma_bridge(c(0, 2, 1.5, 3), c(0, 1, 2, 3), a = 2))

# 9. Errors: the first time must be 0, and "ExtBridge" is not available yet
try(gamma_bridge(x, t + 1, a = 2))
try(gamma_bridge(x, t, a = 2, method = "ExtBridge"))

# 10. Ten trajectories simulated with levysim, inspected more and more often
plan <- levysim::inspection_times("increasing", n = 20, n_traj = 10, tmax = 40)
sims <- levysim::sim_gamma(plan, a = 2, b = 3)

# Parameters estimated with levyfit
a_hat <- levyfit::gamma_mle(sims)["a"]]
r <- gamma_bridge(sims, a = a_hat)
r
plot_bridge(r, trajectory = 1:4)

# Reconstructed increments of all the trajectories
d <- increments(r)
head(d)
# Under the model, the increments follow Gamma(a * teq, b)
ks.test(d$dx, "pgamma", shape = 2 * d$teq[1], rate = 3)
```

`ig_bridge` *Inverse Gaussian bridge: resampling on a regular grid*

Description

Resamples one or several trajectories of a homogeneous inverse Gaussian process, observed at irregular inspection times, onto a regular time grid, by exact simulation of the inverse Gaussian bridge between the known values.

Usage

```
ig_bridge(x, t = NULL, eta, method = "Bridge")
```

Arguments

`x`	Degradation levels of one trajectory; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels (for instance the result of a `levysim` simulation function).
`t`	Inspection times of the trajectory, starting at 0 and strictly increasing. Not used when `x` is a list.
`eta`	Shape parameter $\eta > 0$, for instance the estimate `ig_mle(x)[["eta"]]` of the package `levyfit`.
`method`	`"Bridge"` (default) for the algorithm of Grall-Maës (2012), or `"ExtBridge"` for its extension (not available yet).

Value

An object of class `"levybridge"`: a list with one data frame per trajectory, with columns `t` (regular grid) and `X` (resampled levels). Its attributes are `model`, `parameters`, `method`, `teq` (time step of the grid of each trajectory) and `observed` (the observed data, one data frame per trajectory). Use `increments()` for the reconstructed increments and `plot_bridge()` to compare observed and resampled values.

Model

The homogeneous inverse Gaussian process with parameters $\mu > 0$ and $\eta > 0$ has $X_0 = 0$, independent increments and $X_t - X_s \sim IG(\mu(t-s), \eta(t-s)^2)$ (mean $\mu(t-s)$, shape $\eta(t-s)^2$). Its trajectories are increasing.

Law of the bridge

Let $t_A < t < t_B$, $\tau_1 = t - t_A$, $\tau_2 = t_B - t$, $T = \tau_1 + \tau_2$, $\Delta = x_B - x_A > 0$, $m = \tau_1/\tau_2$ and $\lambda = \eta\tau_1^2/\Delta$. Given $X_{t_A} = x_A$ and $X_{t_B} = x_B$, the ratio $W = (X_t - x_A)/(x_B - X_t)$ ("realized / remaining") is the mixture

$$W \sim IG(m, \lambda) \text{ with probability } \tau_2/T, \qquad W \sim m^2/IG(m, \lambda) \text{ with probability } \tau_1/T,$$

and $X_t = x_A + \Delta\, W/(1+W)$. The law of the bridge does not depend on μ: only η is needed.

The simulation combines this mixture with the method of Michael, Schucany and Haas (1976), with one Gaussian and one uniform variable: with $q = Z^2$, $Z \sim N(0,1)$, the two roots

$$w_2 = m + \frac{m^2 q}{2\lambda} + \frac{m}{2\lambda}\sqrt{4m\lambda q + m^2 q^2}, \qquad w_1 = m^2/w_2,$$

and $W = w_1$ with probability $\tau_1(1+w_1)/(T(m+w_1))$, $W = w_2$ otherwise. The small root is computed as m^2/w_2 to avoid a cancellation. When $\Delta = 0$ (two equal observed values, which can happen after rounding), the bridge is constant.

Resampling algorithm (method "Bridge")

This is the interpolation algorithm of Grall-Maës (2012), used for all the processes of the package. Let $0 = t_0 < t_1 < \cdots < t_N$ be the inspection times of a trajectory and $x_0, \ldots, x_N$ the observed levels. The regular grid is $t'_i = i\,\Delta t_{eq}$, $i = 0, \ldots, N$, with $\Delta t_{eq} = t_N/N$: it has as many points as observations and the same end points.

1. $x'_0 = x_0$ and $x'_N = x_N$.
2. For $i = 1, \ldots, N-1$, in increasing order:
   - if t'_i is an inspection time, x'_i is the observed value;
   - otherwise, x'_i is drawn from the bridge between the nearest known point on the left (an inspection or the grid point t'_{i-1} already drawn, whichever is closer) and the nearest inspection on the right.

By the Markov property of the bridge, the vector $(x'_1, \ldots, x'_{N-1})$ follows exactly the conditional law of $(X_{t'_1}, \ldots, X_{t'_{N-1}})$ given the observations. Hence, if the data come from the model and the parameters are the true ones, the reconstructed increments $x'_i - x'_{i-1}$ (see `increments()`) are independent with the law of $X_{\Delta t_{eq}}$: they can be tested with an i.i.d. goodness-of-fit test. A grid point that equals an inspection time up to rounding errors (relative difference below 10^{-9}) is set to that inspection time.

Method "ExtBridge"

Extension of the method of Grall-Maës that integrates the observations. It is not available yet: `method = "ExtBridge"` stops with a message.

References

Grall-Maës, E. (2012). Use of the Kolmogorov-Smirnov test for gamma process. *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, 226(6), 624-634. doi:10.1177/1748006X12462522

Michael, J. R., Schucany, W. R. and Haas, R. W. (1976). Generating random variates using transformations with multiple roots. *The American Statistician*, 30(2), 88-90. doi:10.1080/00031305.1976.10479147

Ribeiro, C. and Webber, N. (2003). A Monte Carlo method for the normal inverse Gaussian option valuation model using an inverse Gaussian bridge. *Computing in Economics and Finance 2003*, 5.

See Also

- `increments()`: the reconstructed increments;
- `plot_bridge()`: observed and resampled values;
- `gamma_bridge()`: bridge of the gamma process;
- `wiener_bridge()`: bridge of the Wiener process;
- `vg_bridge()`: bridge of the variance-gamma process;
- `meixner_bridge()`: bridge of the Meixner process.

Examples

```
# 1. One trajectory observed at 11 irregular times (mu = 1, eta = 3)
t <- c(0, 1.2, 2, 3.8, 5.5, 7.1, 10.4, 12.9, 15.3, 18.7, 22.5)
x <- c(0, 1.31, 2.05, 3.62, 5.71, 7.15, 10.62, 12.83, 15.02, 18.94, 22.31)
set.seed(1)
r <- ig_bridge(x, t, eta = 3)
r

# The resampled trajectory, on the regular grid 0, 2.25, 4.5, ..., 22.5
r[[1]]

# Time step of the grid: t_N / N = 22.5 / 10
```

```
attr(r, "teq")

# 2. Reconstructed increments, the input of the goodness-of-fit tests
increments(r)

# 3. Observed and resampled values
plot_bridge(r)

# 4. Grid points equal to inspection times keep the observed values:
#    here the grid is 0, 2, 4, 6, 8, 10 and 2 and 6 are inspection times
t2 <- c(0, 2, 3, 6, 7, 10)
x2 <- c(0, 1.4, 2.0, 4.1, 4.6, 6.8)
ig_bridge(x2, t2, eta = 3)[[1]]

# 5. One trajectory given as a data frame (times, then levels)
ig_bridge(data.frame(t = t2, X = x2), eta = 3)

# 6. Several trajectories, each with its own inspection times:
#    one regular grid and one time step per trajectory
liste <- list(data.frame(t = c(0, 1, 3, 6), X = c(0, 0.8, 2.1, 4.3)),
              data.frame(t = c(0, 2, 4, 5, 9), X = c(0, 1.2, 2.9, 3.1, 6.2)))
r2 <- ig_bridge(liste, eta = 3)
r2
attr(r2, "teq")
increments(r2)
plot_bridge(r2, trajectory = 1:2)

# 7. The bridge values are random: set.seed() makes them reproducible
set.seed(2); y1 <- ig_bridge(x, t, eta = 3)[[1]]$X
set.seed(2); y2 <- ig_bridge(x, t, eta = 3)[[1]]$X
identical(y1, y2)

# 8. Equal observed values (after rounding) give a constant bridge,
#    a decrease is impossible for this increasing process
ig_bridge(c(0, 1, 1, 2), c(0, 1, 2.5, 3), eta = 3)[[1]]
try(ig_bridge(c(0, 2, 1.5, 3), c(0, 1, 2, 3), eta = 3))

# 9. Errors: the first time must be 0, and "ExtBridge" is not available yet
try(ig_bridge(x, t + 1, eta = 3))
try(ig_bridge(x, t, eta = 3, method = "ExtBridge"))

# 10. Ten trajectories simulated with levysim, inspected more and more often
plan <- levysim::inspection_times("increasing", n = 20, n_traj = 10, tmax = 40)
sims <- levysim::sim_ig(plan, mu = 1, eta = 3)

# Parameters estimated with levyfit
eta_hat <- levyfit::ig_mle(sims)["eta"]]
r <- ig_bridge(sims, eta = eta_hat)
r
plot_bridge(r, trajectory = 1:4)

# Reconstructed increments of all the trajectories
d <- increments(r)
head(d)
# Under the model, the increments follow IG(mu * teq, eta * teq^2)
# (mean mu * teq, variance mu^3 * teq / eta)
c(mean(d$dx), 1 * d$teq[1], var(d$dx), d$teq[1] / 3)
```

`increments` *Reconstructed increments*

Description

Extracts the increments of the trajectories resampled on the regular grid by a `_bridge` function. Under the model, with the true parameters, these increments are independent and have the law of an increment of the process over a time step `teq` (Grall-Maës, 2012): they are the input of the goodness-of-fit tests.

Usage

```
increments(x)
```

Arguments

`x`	Result of a bridge function of the package, for instance `gamma_bridge()`.

Value

A data frame with one row per increment and three columns:

- `id`: number of the trajectory (1, 2, ...);
- `teq`: time step of the regular grid of this trajectory, t_N/N;
- `dx`: reconstructed increment $x'_i - x'_{i-1}$.

See Also

`gamma_bridge()`, `wiener_bridge()`, `ig_bridge()`, `vg_bridge()`, `meixner_bridge()`

Examples

```
# One gamma trajectory observed at irregular times
t <- c(0, 1.2, 2, 3.8, 5.5, 7.1, 10.4, 12.9, 15.3, 18.7, 22.5)
set.seed(1)
x <- c(0, cumsum(rgamma(10, shape = 2 * diff(t), rate = 3)))
r <- gamma_bridge(x, t, a = 2)
increments(r)

# Two trajectories with different times: one time step per trajectory
liste <- list(data.frame(t = c(0, 1, 3, 6), X = c(0, 0.8, 2.1, 4.3)),
              data.frame(t = c(0, 2, 4, 5, 9), X = c(0, 1.2, 2.9, 3.1, 6.2)))
increments(gamma_bridge(liste, a = 2))
```

`meixner_bridge` *Meixner bridge: resampling on a regular grid*

Description

Resamples one or several trajectories of a Meixner process, observed at irregular inspection times, onto a regular time grid, by exact simulation of the Meixner bridge between the known values.

Usage

```
meixner_bridge(x, t = NULL, a, d, m, method = "Bridge")
```

Arguments

`x`	Degradation levels of one trajectory; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels (for instance the result of a `levysim` simulation function).
`t`	Inspection times of the trajectory, starting at 0 and strictly increasing. Not used when `x` is a list.
`a`	Scale parameter, $a > 0$.
`d`	Shape parameter per unit of time, $d > 0$.
`m`	Location parameter per unit of time, real.
`method`	`"Bridge"` (default) for the algorithm of Grall-Maës (2012), or `"ExtBridge"` for its extension (not available yet).

Value

An object of class `"levybridge"`: a list with one data frame per trajectory, with columns `t` (regular grid) and `X` (resampled levels). Its attributes are `model`, `parameters`, `method`, `teq` (time step of the grid of each trajectory) and `observed` (the observed data, one data frame per trajectory). Use `increments()` for the reconstructed increments and `plot_bridge()` to compare observed and resampled values.

Model

The $Meixner(a, b, d, m)$ distribution, $a > 0$, $-\pi < b < \pi$, $d > 0$, m real, has density

$$f(x) = \frac{(2\cos(b/2))^{2d}}{2a\pi\,\Gamma(2d)} \exp\Big(\frac{b(x-m)}{a}\Big)\Big|\Gamma\Big(d + i\frac{x-m}{a}\Big)\Big|^2.$$

The Meixner process has $X_0 = 0$, independent increments and $X_t - X_s \sim Meixner(a, b, d(t-s), m(t-s))$. Its trajectories are not monotone.

Law of the bridge

Let $t_A < t < t_B$, $h_\alpha(y) = |\Gamma(\alpha + iy)|^2$,

$$\alpha_1 = d(t - t_A), \qquad \alpha_2 = d(t_B - t), \qquad z = \frac{(x_B - x_A) - m(t_B - t_A)}{a}.$$

Given $X_{t_A} = x_A$ and $X_{t_B} = x_B$, $X_t = x_A + m(t - t_A) + aY$, where Y has the density

$$q(y) = \frac{h_{\alpha_1}(y)\, h_{\alpha_2}(z - y)}{2\pi B(2\alpha_1, 2\alpha_2)\, h_{\alpha_1+\alpha_2}(z)}.$$

The law of the bridge does not depend on b: only a, d and m are needed.

Simulation of the bridge

Y is drawn exactly by acceptance-rejection with a Cauchy envelope (algorithm 6 of Tito and Paroissin). For $\alpha > 0$ and y real, $h_\alpha(y) \le \Gamma(\alpha)^2$ and $h_\alpha(y) \le \Gamma(\alpha)^2\alpha^2/(\alpha^2 + y^2)$. With $(y_0, s) = (0, \alpha_1)$ if $\alpha_1 \le \alpha_2$ and $(y_0, s) = (z, \alpha_2)$ otherwise, this gives, for all y,

$$h_{\alpha_1}(y)\, h_{\alpha_2}(z - y) \le \Gamma(\alpha_1)^2\Gamma(\alpha_2)^2 \frac{s^2}{s^2 + (y - y_0)^2},$$

a multiple of the Cauchy density of centre y_0 and scale s. The algorithm repeats, with U and V independent uniform variables on $(0, 1)$:

1. $Y = y_0 + s\tan(\pi(V - 1/2))$, a draw of the Cauchy law;
2. accept Y if
$$U\,\Gamma(\alpha_1)^2\Gamma(\alpha_2)^2 \frac{s^2}{s^2 + (Y - y_0)^2} \le h_{\alpha_1}(Y)\, h_{\alpha_2}(z - Y),$$
otherwise draw again.

The number of draws follows a geometric law with parameter

$$P_{acc} = \frac{2B(2\alpha_1, 2\alpha_2)\, h_{\alpha_1+\alpha_2}(z)}{s\,\Gamma(\alpha_1)^2\Gamma(\alpha_2)^2}.$$

P_{acc} decreases quickly when $|z|$ increases (for $\alpha_1 = \alpha_2 = 1$: 1/3 if $z = 0$, 0.04 if $z = 2$, $5 \cdot 10^{-4}$ if $z = 4$): an increment far from its mean $m(t_B - t_A)$ can need many draws. The function $\log h_\alpha$ is computed with `gsl::lngamma_complex()`.

Resampling algorithm (method "Bridge")

This is the interpolation algorithm of Grall-Maës (2012), used for all the processes of the package. Let $0 = t_0 < t_1 < \cdots < t_N$ be the inspection times of a trajectory and $x_0, \ldots, x_N$ the observed levels. The regular grid is $t'_i = i\,\Delta t_{eq}$, $i = 0, \ldots, N$, with $\Delta t_{eq} = t_N/N$: it has as many points as observations and the same end points.

1. $x'_0 = x_0$ and $x'_N = x_N$.
2. For $i = 1, \ldots, N - 1$, in increasing order:
   - if t'_i is an inspection time, x'_i is the observed value;
   - otherwise, x'_i is drawn from the bridge between the nearest known point on the left (an inspection or the grid point t'_{i-1} already drawn, whichever is closer) and the nearest inspection on the right.

By the Markov property of the bridge, the vector $(x'_1, \ldots, x'_{N-1})$ follows exactly the conditional law of $(X_{t'_1}, \ldots, X_{t'_{N-1}})$ given the observations. Hence, if the data come from the model and the parameters are the true ones, the reconstructed increments $x'_i - x'_{i-1}$ (see `increments()`) are independent with the law of $X_{\Delta t_{eq}}$: they can be tested with an i.i.d. goodness-of-fit test. A grid point that equals an inspection time up to rounding errors (relative difference below 10^{-9}) is set to that inspection time.

Method "ExtBridge"

Extension of the method of Grall-Maës that integrates the observations. It is not available yet: `method = "ExtBridge"` stops with a message.

References

Grall-Maës, E. (2012). Use of the Kolmogorov-Smirnov test for gamma process. *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, 226(6), 624-634. doi:10.1177/1748006X12462522

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Tito, A. R. and Paroissin, C. (in preparation).

See Also

- `increments()`: the reconstructed increments;
- `plot_bridge()`: observed and resampled values;
- `gamma_bridge()`: bridge of the gamma process;
- `wiener_bridge()`: bridge of the Wiener process;
- `ig_bridge()`: bridge of the inverse Gaussian process;
- `vg_bridge()`: bridge of the variance-gamma process.

Examples

```
# 1. One trajectory observed at 11 irregular times (a = 1, b = 0.3, d = 1.5, m = 0.2)
t <- c(0, 1.2, 2, 3.8, 5.5, 7.1, 10.4, 12.9, 15.3, 18.7, 22.5)
x <- c(0, 0.9, 0.4, 2.8, 3.1, 5.6, 7.2, 6.5, 9.8, 11.3, 12.1)
set.seed(1)
r <- meixner_bridge(x, t, a = 1, d = 1.5, m = 0.2)
r

# The resampled trajectory, on the regular grid 0, 2.25, 4.5, ..., 22.5
r[[1]]

# Time step of the grid: t_N / N = 22.5 / 10
attr(r, "teq")

# 2. Reconstructed increments, the input of the goodness-of-fit tests
increments(r)

# 3. Observed and resampled values
plot_bridge(r)

# 4. Grid points equal to inspection times keep the observed values:
#    here the grid is 0, 2, 4, 6, 8, 10 and 2 and 6 are inspection times
t2 <- c(0, 2, 3, 6, 7, 10)
x2 <- c(0, 1.4, 0.9, 3.1, 2.6, 4.8)
meixner_bridge(x2, t2, a = 1, d = 1.5, m = 0.2)[[1]]

# 5. One trajectory given as a data frame (times, then levels)
meixner_bridge(data.frame(t = t2, X = x2), a = 1, d = 1.5, m = 0.2)
```

```
# 6. Several trajectories, each with its own inspection times:
#    one regular grid and one time step per trajectory
liste <- list(data.frame(t = c(0, 1, 3, 6), X = c(0, 0.4, 2.3, 1.9)),
              data.frame(t = c(0, 2, 4, 5, 9), X = c(0, 1.6, 0.8, 3.1, 4.2)))
r2 <- meixner_bridge(liste, a = 1, d = 1.5, m = 0.2)
r2
attr(r2, "teq")
increments(r2)
plot_bridge(r2, trajectory = 1:2)

# 7. The bridge values are random: set.seed() makes them reproducible
set.seed(2); y1 <- meixner_bridge(x, t, a = 1, d = 1.5, m = 0.2)[[1]]$X
set.seed(2); y2 <- meixner_bridge(x, t, a = 1, d = 1.5, m = 0.2)[[1]]$X
identical(y1, y2)

# 8. Errors: the first time must be 0, and "ExtBridge" is not available yet
try(meixner_bridge(x, t + 1, a = 1, d = 1.5, m = 0.2))
try(meixner_bridge(x, t, a = 1, d = 1.5, m = 0.2, method = "ExtBridge"))

# 9. Ten trajectories simulated with levysim, inspected more and more often
plan <- levysim::inspection_times("increasing", n = 20, n_traj = 10, tmax = 40)
sims <- levysim::sim_meixner(plan, a = 1, b = 0.3, d = 1.5, m = 0.2)

# With the true parameters (b is not needed by the bridge)
r <- meixner_bridge(sims, a = 1, d = 1.5, m = 0.2)
r
plot_bridge(r, trajectory = 1:4)

# Reconstructed increments of all the trajectories
d <- increments(r)
head(d)
# Under the model, the increments have mean (m + a d tan(b/2)) teq
# and variance a^2 d teq / (2 cos(b/2)^2)
c(mean(d$dx), (0.2 + 1.5 * tan(0.15)) * d$teq[1])
c(var(d$dx), 1.5 * d$teq[1] / (2 * cos(0.15)^2))
```

`plot_bridge` *Plot observed and resampled trajectories*

Description

Draws, for one or several trajectories, the observed values at the inspection times and the values resampled on the regular grid by a `_bridge` function, with `ggplot2`. The plot shows how the bridge fills the gaps between the inspections.

Usage

```
plot_bridge(x, trajectory = 1, title = NULL)
```

Arguments

`x`	Result of a bridge function of the package, for instance `gamma_bridge()`.

`trajectory`	Numbers of the trajectories to plot, for instance `1` or `1:3` (default: the first one). With several trajectories, one panel is drawn per trajectory.
`title`	Title of the plot, a character string; by default the name of the bridge, for instance `"gamma bridge"`.

Value

A `ggplot` object. It is displayed when printed, and it can be modified with the usual `ggplot2` functions, added with `+`: for instance `+ ggplot2::labs(x = "Time (hours)")`.

See Also

`increments()`

Examples

```
t <- c(0, 1.2, 2, 3.8, 5.5, 7.1, 10.4, 12.9, 15.3, 18.7, 22.5)
set.seed(1)
x <- c(0, cumsum(rgamma(10, shape = 2 * diff(t), rate = 3)))
r <- gamma_bridge(x, t, a = 2)

# Observed values (red) and resampled values on the regular grid (blue)
plot_bridge(r)

# Custom title and axis labels
plot_bridge(r, title = "Gamma bridge, a = 2") +
  ggplot2::labs(x = "Time (hours)", y = "Degradation (%)")

# Several trajectories: one panel per trajectory
liste <- list(data.frame(t = c(0, 1, 3, 6), X = c(0, 0.8, 2.1, 4.3)),
              data.frame(t = c(0, 2, 4, 5, 9), X = c(0, 1.2, 2.9, 3.1, 6.2)))
plot_bridge(gamma_bridge(liste, a = 2), trajectory = 1:2)
```

---

vg_bridge *Variance-gamma bridge: resampling on a regular grid*

Description

Resamples one or several trajectories of a variance-gamma process, observed at irregular inspection times, onto a regular time grid, by exact simulation of the variance-gamma bridge between the known values.

Usage

```
vg_bridge(x, t = NULL, theta, sigma, nu, method = "Bridge")
```

Arguments

`x`	Degradation levels of one trajectory; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels (for instance the result of a `levysim` simulation function).
`t`	Inspection times of the trajectory, starting at 0 and strictly increasing. Not used when `x` is a list.

`theta`	Drift of the subordinated Brownian motion, real.
`sigma`	Volatility of the subordinated Brownian motion, $\sigma > 0$.
`nu`	Variance rate of the gamma clock, $\nu > 0$.
`method`	`"Bridge"` (default) for the algorithm of Grall-Maës (2012), or `"ExtBridge"` for its extension (not available yet).

Value

An object of class `"levybridge"`: a list with one data frame per trajectory, with columns `t` (regular grid) and `X` (resampled levels). Its attributes are `model`, `parameters`, `method`, `teq` (time step of the grid of each trajectory) and `observed` (the observed data, one data frame per trajectory). Use `increments()` for the reconstructed increments and `plot_bridge()` to compare observed and resampled values.

Model

The variance-gamma process with parameters θ, $\sigma > 0$ and $\nu > 0$ is $X_s = \theta\gamma_s + \sigma W_{\gamma_s}$, where γ is a gamma process with $\gamma_s \sim \Gamma(s/\nu)$ with scale ν (mean s, variance νs) and W an independent standard Brownian motion. Its trajectories are not monotone.

Law of the bridge

Let $t_A < t < t_B$, $\tau_1 = t - t_A$, $\tau_2 = t_B - t$, $T = \tau_1 + \tau_2$, $\Delta = x_B - x_A$,

$$p = \frac{T}{\nu} - \frac{1}{2}, \qquad \kappa^2 = \frac{\theta^2 + 2\sigma^2/\nu}{\sigma^4}.$$

Given $X_{t_A} = x_A$ and $X_{t_B} = x_B$,

$$X_t = x_A + Y\Delta + \sqrt{S\,Y(1-Y)}\,\xi,$$

with three independent variables $S \sim \mathrm{GIG}(p, \kappa^2, \Delta^2)$ (density proportional to $s^{p-1}\exp(-(\kappa^2 s + \Delta^2/s)/2)$), $Y \sim \mathrm{Beta}(\tau_1/\nu, \tau_2/\nu)$ and $\xi \sim N(0,1)$. S/σ^2 is the increment of the clock γ over $[t_A, t_B]$, Y the fraction of it reached at t, and, given the clock, X is a Brownian bridge. The law of the bridge depends on (θ, σ, ν) only through ν and κ. The GIG variable is drawn with `GIGrvg::rgig()`. The case $\Delta = 0$ with $p \le 0$, which has probability zero, gives $S = 0$.

Resampling algorithm (method "Bridge")

This is the interpolation algorithm of Grall-Maës (2012), used for all the processes of the package. Let $0 = t_0 < t_1 < \cdots < t_N$ be the inspection times of a trajectory and $x_0, \ldots, x_N$ the observed levels. The regular grid is $t'_i = i\,\Delta t_{eq}$, $i = 0, \ldots, N$, with $\Delta t_{eq} = t_N/N$: it has as many points as observations and the same end points.

1. $x'_0 = x_0$ and $x'_N = x_N$.
2. For $i = 1, \ldots, N-1$, in increasing order:
   - if t'_i is an inspection time, x'_i is the observed value;
   - otherwise, x'_i is drawn from the bridge between the nearest known point on the left (an inspection or the grid point t'_{i-1} already drawn, whichever is closer) and the nearest inspection on the right.

By the Markov property of the bridge, the vector $(x'_1, \ldots, x'_{N-1})$ follows exactly the conditional law of $(X_{t'_1}, \ldots, X_{t'_{N-1}})$ given the observations. Hence, if the data come from the model and the parameters are the true ones, the reconstructed increments $x'_i - x'_{i-1}$ (see `increments()`) are independent with the law of $X_{\Delta t_{eq}}$: they can be tested with an i.i.d. goodness-of-fit test. A grid point that equals an inspection time up to rounding errors (relative difference below 10^{-9}) is set to that inspection time.

Method "ExtBridge"

Extension of the method of Grall-Maës that integrates the observations. It is not available yet: `method = "ExtBridge"` stops with a message.

References

Grall-Maës, E. (2012). Use of the Kolmogorov-Smirnov test for gamma process. *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, 226(6), 624-634. doi:10.1177/1748006X12462522

Madan, D. B., Carr, P. P. and Chang, E. C. (1998). The variance gamma process and option pricing. *European Finance Review*, 2(1), 79-105. doi:10.1023/A:1009703431535

Hörmann, W. and Leydold, J. (2014). Generating generalized inverse Gaussian random variates. *Statistics and Computing*, 24, 547-557. doi:10.1007/s1122201393873

See Also

- `increments()`: the reconstructed increments;
- `plot_bridge()`: observed and resampled values;
- `gamma_bridge()`: bridge of the gamma process;
- `wiener_bridge()`: bridge of the Wiener process;
- `ig_bridge()`: bridge of the inverse Gaussian process;
- `meixner_bridge()`: bridge of the Meixner process.

Examples

```
# 1. One trajectory observed at 11 irregular times (theta = 0.5, sigma = 1.5, nu = 2)
t <- c(0, 1.2, 2, 3.8, 5.5, 7.1, 10.4, 12.9, 15.3, 18.7, 22.5)
x <- c(0, 0.9, 0.4, 2.8, 3.1, 5.6, 7.2, 6.5, 9.8, 11.3, 12.1)
set.seed(1)
r <- vg_bridge(x, t, theta = 0.5, sigma = 1.5, nu = 2)
r

# The resampled trajectory, on the regular grid 0, 2.25, 4.5, ..., 22.5
r[[1]]

# Time step of the grid: t_N / N = 22.5 / 10
attr(r, "teq")

# 2. Reconstructed increments, the input of the goodness-of-fit tests
increments(r)

# 3. Observed and resampled values
plot_bridge(r)

# 4. Grid points equal to inspection times keep the observed values:
```

```
#    here the grid is 0, 2, 4, 6, 8, 10 and 2 and 6 are inspection times
t2 <- c(0, 2, 3, 6, 7, 10)
x2 <- c(0, 1.4, 0.9, 3.1, 2.6, 4.8)
vg_bridge(x2, t2, theta = 0.5, sigma = 1.5, nu = 2)[[1]]

# 5. One trajectory given as a data frame (times, then levels)
vg_bridge(data.frame(t = t2, X = x2), theta = 0.5, sigma = 1.5, nu = 2)

# 6. Several trajectories, each with its own inspection times:
#    one regular grid and one time step per trajectory
liste <- list(data.frame(t = c(0, 1, 3, 6), X = c(0, 0.4, 2.3, 1.9)),
              data.frame(t = c(0, 2, 4, 5, 9), X = c(0, 1.6, 0.8, 3.1, 4.2)))
r2 <- vg_bridge(liste, theta = 0.5, sigma = 1.5, nu = 2)
r2
attr(r2, "teq")
increments(r2)
plot_bridge(r2, trajectory = 1:2)

# 7. The bridge values are random: set.seed() makes them reproducible
set.seed(2); y1 <- vg_bridge(x, t, theta = 0.5, sigma = 1.5, nu = 2)[[1]]$X
set.seed(2); y2 <- vg_bridge(x, t, theta = 0.5, sigma = 1.5, nu = 2)[[1]]$X
identical(y1, y2)

# 8. Errors: the first time must be 0, and "ExtBridge" is not available yet
try(vg_bridge(x, t + 1, theta = 0.5, sigma = 1.5, nu = 2))
try(vg_bridge(x, t, theta = 0.5, sigma = 1.5, nu = 2, method = "ExtBridge"))

# 9. Ten trajectories simulated with levysim, inspected more and more often
plan <- levysim::inspection_times("increasing", n = 20, n_traj = 10, tmax = 40)
sims <- levysim::sim_vg(plan, theta = 0.5, sigma = 1.5, nu = 2)

# Parameters estimated with levyfit
# With the true parameters (they can be estimated with levyfit::vg_mle())
r <- vg_bridge(sims, theta = 0.5, sigma = 1.5, nu = 2)
r
plot_bridge(r, trajectory = 1:4)

# Reconstructed increments of all the trajectories
d <- increments(r)
head(d)
# Under the model, the increments have mean theta * teq and
# variance (sigma^2 + nu * theta^2) * teq
c(mean(d$dx), 0.5 * d$teq[1], var(d$dx), (1.5^2 + 2 * 0.5^2) * d$teq[1])
```

`wiener_bridge` *Brownian bridge: resampling on a regular grid*

Description

Resamples one or several trajectories of a Wiener process with drift, observed at irregular inspection times, onto a regular time grid, by exact simulation of the Brownian bridge between the known values.

Usage

```
wiener_bridge(x, t = NULL, sigma, method = "Bridge")
```

Arguments

`x`	Degradation levels of one trajectory; or a list of data frames, one per trajectory, whose first column contains the inspection times and second column the degradation levels (for instance the result of a `levysim` simulation function).
`t`	Inspection times of the trajectory, starting at 0 and strictly increasing. Not used when `x` is a list.
`sigma`	Volatility $\sigma > 0$ (standard deviation per square root of time unit). The package `levyfit` returns the variance: use `sqrt(wiener_mle(x)[["sigma2"]])`.
`method`	`"Bridge"` (default) for the algorithm of Grall-Maës (2012), or `"ExtBridge"` for its extension (not available yet).

Value

An object of class `"levybridge"`: a list with one data frame per trajectory, with columns `t` (regular grid) and `X` (resampled levels). Its attributes are `model`, `parameters`, `method`, `teq` (time step of the grid of each trajectory) and `observed` (the observed data, one data frame per trajectory). Use `increments()` for the reconstructed increments and `plot_bridge()` to compare observed and resampled values.

Model

The Wiener process with drift μ and volatility $\sigma > 0$ is $X_t = \mu t + \sigma B_t$, with B a standard Brownian motion: $X_0 = 0$, independent increments and $X_t - X_s \sim N(\mu(t-s), \sigma^2(t-s))$. Its trajectories are not monotone.

Law of the bridge

Let $t_A < t < t_B$, $\tau_1 = t - t_A$, $\tau_2 = t_B - t$, $T = \tau_1 + \tau_2$ and $\Delta = x_B - x_A$. Given $X_{t_A} = x_A$ and $X_{t_B} = x_B$,

$$X_t \sim N\Big(x_A + \Delta\frac{\tau_1}{T},\ \sigma^2\frac{\tau_1\tau_2}{T}\Big).$$

The law of the bridge does not depend on the drift μ: only σ is needed.

Resampling algorithm (method "Bridge")

This is the interpolation algorithm of Grall-Maës (2012), used for all the processes of the package. Let $0 = t_0 < t_1 < \cdots < t_N$ be the inspection times of a trajectory and $x_0, \ldots, x_N$ the observed levels. The regular grid is $t'_i = i\,\Delta t_{eq}$, $i = 0, \ldots, N$, with $\Delta t_{eq} = t_N/N$: it has as many points as observations and the same end points.

1. $x'_0 = x_0$ and $x'_N = x_N$.
2. For $i = 1, \ldots, N-1$, in increasing order:
   - if t'_i is an inspection time, x'_i is the observed value;
   - otherwise, x'_i is drawn from the bridge between the nearest known point on the left (an inspection or the grid point t'_{i-1} already drawn, whichever is closer) and the nearest inspection on the right.

By the Markov property of the bridge, the vector $(x'_1, \ldots, x'_{N-1})$ follows exactly the conditional law of $(X_{t'_1}, \ldots, X_{t'_{N-1}})$ given the observations. Hence, if the data come from the model and the parameters are the true ones, the reconstructed increments $x'_i - x'_{i-1}$ (see `increments()`) are independent with the law of $X_{\Delta t_{eq}}$: they can be tested with an i.i.d. goodness-of-fit test. A grid point that equals an inspection time up to rounding errors (relative difference below 10^{-9}) is set to that inspection time.

Method "ExtBridge"

Extension of the method of Grall-Maës that integrates the observations. It is not available yet: `method = "ExtBridge"` stops with a message.

References

Grall-Maës, E. (2012). Use of the Kolmogorov-Smirnov test for gamma process. *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, 226(6), 624-634. doi:10.1177/1748006X12462522

See Also

- `increments()`: the reconstructed increments;
- `plot_bridge()`: observed and resampled values;
- `gamma_bridge()`: bridge of the gamma process;
- `ig_bridge()`: bridge of the inverse Gaussian process;
- `vg_bridge()`: bridge of the variance-gamma process;
- `meixner_bridge()`: bridge of the Meixner process.

Examples

```
# 1. One trajectory observed at 11 irregular times (mu = 0.5, sigma = 1.5)
t <- c(0, 1.2, 2, 3.8, 5.5, 7.1, 10.4, 12.9, 15.3, 18.7, 22.5)
set.seed(1)
x <- c(0, cumsum(0.5 * diff(t) + 1.5 * sqrt(diff(t)) * rnorm(10)))
set.seed(1)
r <- wiener_bridge(x, t, sigma = 1.5)
r

# The resampled trajectory, on the regular grid 0, 2.25, 4.5, ..., 22.5
r[[1]]

# Time step of the grid: t_N / N = 22.5 / 10
attr(r, "teq")

# 2. Reconstructed increments, the input of the goodness-of-fit tests
increments(r)

# 3. Observed and resampled values
plot_bridge(r)

# 4. Grid points equal to inspection times keep the observed values:
#    here the grid is 0, 2, 4, 6, 8, 10 and 2 and 6 are inspection times
t2 <- c(0, 2, 3, 6, 7, 10)
x2 <- c(0, 1.4, 0.9, 3.1, 2.6, 4.8)
wiener_bridge(x2, t2, sigma = 1.5)[[1]]
```

```
# 5. One trajectory given as a data frame (times, then levels)
wiener_bridge(data.frame(t = t2, X = x2), sigma = 1.5)

# 6. Several trajectories, each with its own inspection times:
#    one regular grid and one time step per trajectory
liste <- list(data.frame(t = c(0, 1, 3, 6), X = c(0, 0.4, 2.3, 1.9)),
              data.frame(t = c(0, 2, 4, 5, 9), X = c(0, 1.6, 0.8, 3.1, 4.2)))
r2 <- wiener_bridge(liste, sigma = 1.5)
r2
attr(r2, "teq")
increments(r2)
plot_bridge(r2, trajectory = 1:2)

# 7. The bridge values are random: set.seed() makes them reproducible
set.seed(2); y1 <- wiener_bridge(x, t, sigma = 1.5)[[1]]$X
set.seed(2); y2 <- wiener_bridge(x, t, sigma = 1.5)[[1]]$X
identical(y1, y2)

# 8. Errors: the first time must be 0, and "ExtBridge" is not available yet
try(wiener_bridge(x, t + 1, sigma = 1.5))
try(wiener_bridge(x, t, sigma = 1.5, method = "ExtBridge"))

# 9. Ten trajectories simulated with levysim, inspected more and more often
plan <- levysim::inspection_times("increasing", n = 20, n_traj = 10, tmax = 40)
sims <- levysim::sim_wiener(plan, mu = 0.5, sigma = 1.5)

# Parameters estimated with levyfit
sigma_hat <- sqrt(levyfit::wiener_mle(sims)[["sigma2"]])
r <- wiener_bridge(sims, sigma = sigma_hat)
r
plot_bridge(r, trajectory = 1:4)

# Reconstructed increments of all the trajectories
d <- increments(r)
head(d)
# Under the model, the increments follow N(mu * teq, sigma^2 * teq)
ks.test(d$dx, "pnorm", mean = 0.5 * d$teq[1], sd = 1.5 * sqrt(d$teq[1]))
```

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